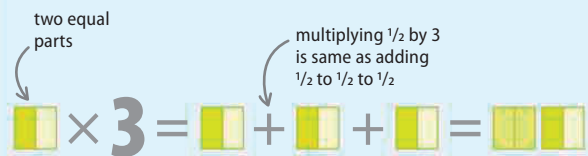


MULTIPLYING FRACTIONS

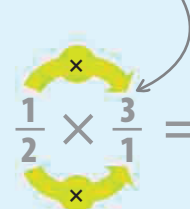
Fractions can be multiplied by other fractions. To multiply fractions by mixed fractions or whole numbers, they first need to be converted into improper (top-heavy) fractions.



$$\frac{1}{2} \times 3 = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 1\frac{1}{2}$$

Imagine multiplying a fraction by a whole number as adding the fraction to itself that many times. Alternatively, imagine multiplying a whole number by a fraction as taking that portion of the whole number, here $\frac{1}{2}$ of 3.

whole number an improper fraction with whole number as numerator and 1 as denominator



Convert the whole number to a fraction. Next, multiply both numerators together and then both denominators.

remainder becomes numerator of fraction

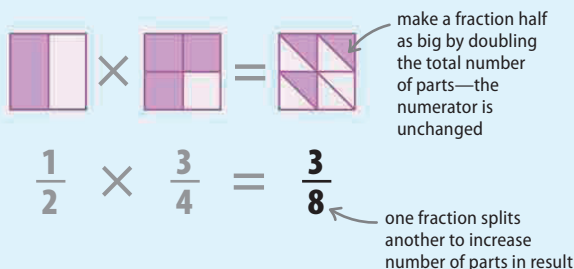
$$\frac{1}{2} \times \frac{3}{1} = \frac{3}{2} \rightarrow 3 \div 2 = 1\text{r}1 = 1\frac{1}{2}$$

denominator stays the same

Divide the numerator of the resulting fraction by the denominator. The answer is given as a mixed fraction.

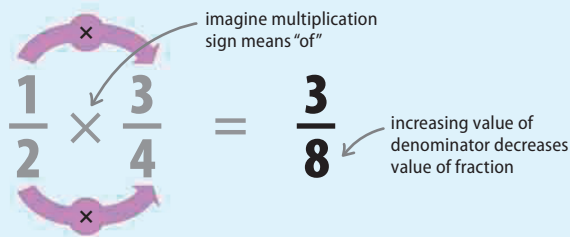
Multiplying two proper fractions

Proper fractions can be multiplied by each other. It is useful to imagine that the multiplication sign means “of”—the problem below can be expressed as “what is $\frac{1}{2}$ of $\frac{3}{4}$?” and “what is $\frac{3}{4}$ of $\frac{1}{2}$?”.



$$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$$

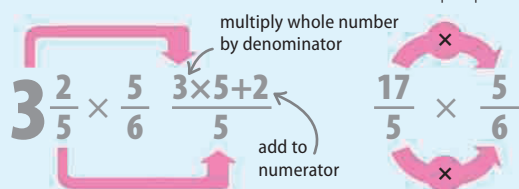
Visually, the result of multiplying two proper fractions is that the space taken by both together is reduced.



Multiply the numerators and the denominators. The resulting fraction answers both questions: “what is $\frac{1}{2}$ of $\frac{3}{4}$?” and “what is $\frac{3}{4}$ of $\frac{1}{2}$?”.

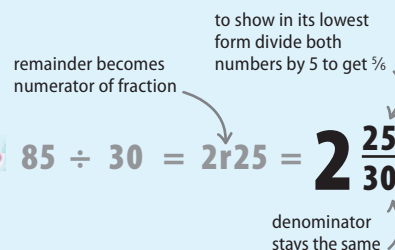
Multiplying mixed fractions

To multiply a proper fraction by a mixed fraction, it is necessary to first convert the mixed fraction into an improper fraction.



First, turn the mixed fraction into an improper fraction.

Next, multiply the numerators and denominators of both fractions to get a new fraction.



Divide the numerator of the new improper fraction by its denominator. The answer is shown as a mixed fraction.